

Semantic Geometry and Symbolic Load: Encoding Polygonal Typologies in AI Prompt Architectures for Visual-Symbolic Influence

Introduction: Framing Polygonal Cognition in Generative Systems

The rapid proliferation of generative Artificial Intelligence (AI) systems, particularly large language models (LLMs) and diffusion models, has revolutionized content creation across various modalities.¹ These systems demonstrate remarkable capabilities in generating novel text, images, music, and even motion sequences.³ However, achieving precise control over the semantic content, compositional structure, and aesthetic qualities of generated outputs remains a significant challenge. While prompt engineering offers a primary interface for user guidance², the effective vocabulary for influencing deeper structural and symbolic aspects is still evolving.

Within this context, geometric primitives represent an underexplored yet potentially powerful set of control elements.⁵ Regular polygons, ranging from the fundamental triangle (3-gon) to complex, near-circular high-n shapes (e.g., 100-gon), possess inherent structural properties (symmetry, side count, angular relationships) that have historically accrued symbolic meaning across diverse cultural and scientific domains.⁶ This report investigates the concept of "Polygonal Cognition" in generative systems, examining how these geometric forms can function as latent symbolic grammars within AI prompt architectures.

The central problem addressed is twofold: first, understanding how the structural and symbolic affordances of polygons, from simple to highly complex, can be computationally encoded and interpreted by current generative models (primarily LLMs and diffusion models); second, determining how this encoding can be leveraged to guide outputs towards coherent, symbolically meaningful, and compositionally stable renderings across visual, musical, and motion domains. This exploration necessitates navigating the intricate interplay between the formal properties of geometry, the fluid nature of symbolism, the architectural limitations of AI models, and the constraints of human perception.

To structure this inquiry, several theoretical lenses are employed. Computational semiotics provides frameworks for analyzing how signs (including geometric shapes) acquire and communicate meaning within systems, both human and artificial.⁸ Concepts from computational grammar inform the analysis of prompt structures and the potential for formalizing geometric encoding. Generative design theory offers perspectives on using algorithmic processes to create and manipulate forms.¹⁰ Finally, insights from visual and cognitive psychology regarding the perception of shape, complexity, and aesthetic harmony ground the discussion in human factors.¹²

This report systematically examines polygonal symbolism and its cultural drift, explores methods for encoding geometry in prompts and the associated challenges of AI tokenization and latent representation, analyzes the perceptual and cognitive limits influencing both human and AI interpretation, catalogues common failure modes in generative fidelity, and investigates the potential for transferring polygonal encoding across different modalities. The ultimate aim is to provide a theoretically grounded yet practically relevant understanding of how semantic geometry can be harnessed—and where it currently fails—within the complex landscape of AI-driven multimodal composition.

I. The Symbolic Geometry of Polygons: Affordance, Complexity, and Cultural Drift

The potential for polygons to function as symbolic elements within generative systems stems from their fundamental geometric properties and the layers of meaning accrued through cultural usage and psychological association. Understanding this relationship requires examining both the formal characteristics of polygons and the dynamic nature of their symbolism.

1.1. Foundational Geometric Properties

A regular polygon is defined in Euclidean geometry as being both equiangular (all interior angles equal) and equilateral (all sides of equal length).⁶ Regular polygons can be convex (like a standard square or hexagon) or star-shaped (like a pentagram, denoted with Schläfli symbols such as $\{5/2\}$).⁶ Key properties relevant to their potential symbolic function include:

- **Symmetry:** An n -sided regular polygon possesses a high degree of symmetry, described by the dihedral group D_n of order $2n$. This group includes n rotational symmetries and n reflectional symmetries passing through the center.⁶ This inherent order and balance are foundational to many symbolic interpretations related to stability and harmony.
- **Cyclic and Tangential Nature:** Every regular polygon can be inscribed in a circle (cyclic) and has an incircle tangent to each side midpoint (tangential).⁶ This relationship to the circle, a potent symbol itself, becomes particularly relevant as the number of sides increases.
- **Angle Measures:** The interior angle of a regular n -gon is given by $180(n-2)/n$ degrees, while the exterior angle is $360/n$ degrees.¹⁵ These predictable angular relationships contribute to their structural integrity and visual distinctiveness, especially for low values of n .
- **Approximation of Circle:** As the number of sides n increases, a sequence of regular polygons with a fixed perimeter or area visually and mathematically approximates a circle.⁶ This property influences the perception and potential symbolism of high- n polygons.

These formal geometric properties provide the substrate upon which symbolic meanings are built. Symmetry relates to balance and order, side count relates to numerical symbolism, and the relationship to the circle informs ideas of completeness or diffusion.

1.2. Symbolic Associations: Low- n Polygons (3-gon to ~12-gon)

Simple polygons with fewer sides have accumulated rich and often varied symbolic associations across history and cultures:

- **Triangle (3-gon):** Frequently associated with trinities (religious, philosophical), directionality (arrows), strength, stability (when base-down), dynamism, and potential conflict.⁷ Its minimal structure makes it a fundamental building block.
- **Square (4-gon):** Commonly symbolizes stability, reliability, order, limitation, structure, and groundedness, often linked to the built environment and concepts of fairness (a "square deal").⁷
- **Pentagon (5-gon):** Associated with the five elements in some traditions (earth, water, air, fire, spirit)⁷, the human form (via proportions related to the golden ratio, though not explicitly stated in snippets), and notions of harmony. The pentagram ($\{5/2\}$) variant carries complex historical weight, representing mathematical perfection (Pythagoreans), the five wounds of Christ in early Christianity, the seal of Jerusalem in Judaism, protection in witchcraft, and later, through inversion and cultural shifts, associations with evil.¹⁴ It has also been linked to national power and prophecy in specific Native American traditions.¹⁸
- **Hexagon (6-gon):** Often symbolizes efficiency, harmony, balance, and natural optimization, famously seen in

beehives¹⁵ and molecular structures like benzene.¹⁹ It represents structure and connection. Culturally, "l'Hexagone" refers to mainland France.¹⁹ Its use in advanced technologies like telescope mirrors suggests precision and innovation.¹⁹

- **Octagon (8-gon):** Can symbolize regeneration, infinity, transition (being between a square and a circle), and totality (e.g., the eight trigrams of the Bagua).⁷ It features prominently in architecture across cultures, including Islamic art (Rub el Hizb motif, Dome of the Rock), Christian churches (San Vitale, Aachen Cathedral's Palatine Chapel), and ancient structures (Tower of the Winds).²⁰ Its shape is also recognized in everyday objects like stop signs.²⁰
- **Decagon (10-gon):** Associated with order, completion, and unity, potentially linked to base-10 numerical systems.²¹ It was considered sacred in ancient Greece and linked to Pythagorean thought.²² Some cultures view its tenfold symmetry as representing completeness or oneness.²¹

The psychological basis for some of these associations appears rooted in fundamental perceptual experiences. Angular shapes like triangles evoke tension or dynamism, while stable shapes like squares suggest reliability.⁷ Symmetrical forms are generally perceived positively, relating to Gestalt principles of order.¹⁶

1.3. Symbolic Scarcity: High-n Polygons (> ~12-gon)

Beyond approximately the dodecagon (12-gon), specific, widely recognized symbolic meanings for individual regular polygons become scarce. An enneacontagon (90-gon), for instance, is primarily defined by its geometric properties – 90 sides and 90 angles – with no readily available historical or cultural symbolism attached.²⁴

Consequently, the primary symbolic affordance of high-n polygons shifts. Instead of representing specific concepts like 'stability' or 'trinity', they tend to symbolize:

- **Complexity:** Their numerous sides and vertices inherently convey intricacy and detail.
- **Approximation:** As n increases, they visually approach a circle, symbolizing convergence, near-completeness, or the boundary before becoming continuous.⁶
- **Diffusion/Multiplicity:** The high number of sides can suggest distribution, numerousness, or a lack of sharp definition.
- **Generality/Abstraction:** They may function as generic representations of 'shape' or 'enclosure' without specific connotations.
- **Potential Overload/Noise:** As suggested in the initial query, extreme values of n might signify excessive complexity bordering on incoherence or perceptual noise.

1.4. Semantic Load Matrix: Quantifying Meaning Complexity

To structure the relationship between polygon complexity and symbolic meaning, a conceptual "Semantic Load Matrix" can be proposed. This matrix maps the number of sides (n) to several dimensions:

Polygon Range (n)	Specificity of Meaning	Cultural Variance	Theorized Generative Stability	Estimated Semantic Load
3-6	High	High	Medium-High	High

7-12	Medium	Medium	High	Medium
13--30	Low	Low	Medium	Low
>30	Very Low	Very Low	Low	Very Low

- **Specificity of Meaning:** Decreases as n increases. Low-n polygons have distinct, albeit sometimes multiple, meanings (e.g., triangle as 'trinity' or 'conflict'). High-n polygons converge on the general meaning of 'complexity' or 'circle-likeness'.
- **Cultural Variance:** High for low-n polygons where different cultures assign diverse or conflicting meanings (e.g., pentagram¹⁷). Low for high-n polygons, where the meaning (or lack thereof) is more consistent.
- **Theorized Generative Stability:** This hypothesizes how easily and reliably a generative model might render the polygon and its associated semantics. It may follow an inverted U-shape, mirroring findings in aesthetic preference for complexity.¹³ Very low-n shapes might suffer from ambiguity or conflicting interpretations in the training data. Mid-range polygons (perhaps 7-12 sides) might hit a sweet spot of distinct structure and sufficient training examples. High-n polygons are expected to have low stability due to representational limits and rendering failures (discussed further in Section IV).
- **Semantic Load:** An overall measure combining specificity and variance. Low-n polygons carry a high load (rich, varied meanings). High-n polygons carry a low load (generic meaning).

This matrix highlights a crucial point for generative systems: effective semantic communication relies on shared understanding or a common knowledge base between transmitter and receiver (or prompter and AI).³ High-n polygons lack this established, shared symbolic knowledge base, making their specific semantic control potential weak, defaulting instead to conveying 'complexity'.

1.5. Cultural and Contextual Drift

The symbolic meaning of any sign, including geometric shapes, is not fixed but drifts across time, culture, and context.⁸ The pentagram's journey from Pythagorean perfection to Christian symbol to occult signifier and perceived mark of evil is a prime example.¹⁷ The color white signifies purity in some Western cultures but death in parts of East Asia.²⁸ Similarly, a hexagon's meaning shifts depending on whether it's encountered in a beehive (nature, efficiency), a benzene diagram (science, structure), or as a national identifier (culture, France).¹⁹ Sacred geometry itself shows evolution, used differently by Egyptians, Greeks, and later spiritual traditions.²⁹

This drift extends beyond historical and cultural dimensions. The meaning associated with a polygon can vary significantly across domains like science, mythology, art, and everyday life. Furthermore, the advent of generative AI introduces a new layer of potential drift. AI models are trained on vast datasets encompassing diverse cultural artifacts, scientific diagrams, artistic representations, and textual descriptions.¹ The statistical patterns learned by the model may lead it to associate polygons with meanings that are an unpredictable amalgamation of these varied inputs, potentially diverging from any single cultural or historical interpretation. The model's internal "understanding" of a polygon's symbolism is thus mediated by the biases and composition of its training data, creating a form of computationally-mediated semantic drift. The meaning evoked by prompting "hexagon" might be subtly different in an AI trained predominantly on scientific papers versus one trained on art history archives.

(See Table 1: Shape-to-Meaning Lexicon at the end of this section)

Table 1: Shape-to-Meaning Lexicon (Selected Examples)

Polygon Name	Sides (n)	Key Geometric Properties (Symmetry: Dn)	Established Symbolic Associations (Contextual Notes)	Theorized High-n Affordances	Estimated Semantic Load	Potential Generative Effect
Triangle	3	D3, Convex	Trinity, directionality, stability (base-down), dynamism, conflict, strength ⁷	N/A	High	Dynamic, Stable (contextual)
Square	4	D4, Convex	Stability, order, reliability, limitation, equality, earthliness ⁷	N/A	High	Stable, Grounded
Pentagon	5	D5, Convex/Star ({5/2})	Elements, human form, harmony ⁷ ; Pentagram: perfection, protection, occult significance, potential 'evil' inversion ¹⁴ ; National power (specific Native American tradition) ¹⁸	N/A	High	Balanced, Potentially Tense
Hexagon	6	D6, Convex	Efficiency, harmony, natural optimization (bees ¹⁵), structure (chemistry), connection, community; National identity (France ¹⁹)	N/A	Medium	Stable, Structured, Efficient
Octagon	8	D8, Convex	Regeneration, infinity,	N/A	Medium	Balanced, Transitional

			transition, totality (Bagua ²⁰); Architectural prevalence (Islamic, Christian ²⁰); Stop sign shape ²⁰			
Decagon	10	D10, Convex	Order cycles, completeness, oneness, tenfold symmetry ²¹ ; Sacred geometry (Ancient Greece ²²)	N/A	Medium	Ordered, Complete
Dodecagon	12	D12, Convex	Cycles (zodiac, time), completeness	N/A	Low-Medium	Ordered, Cyclical
Icosagon	20	D20, Convex	(Few specific associations)	Complexity, Approximation	Low	Complex, Near-Circular
Triacontagon	30	D30, Convex	(No specific associations)	Complexity, Approximation	Very Low	Complex, Circular, Unstable?
Enneacontagon	90	D90, Convex	(No specific associations beyond geometry ²⁴)	Complexity, Diffusion, Overload	Very Low	Diffuse, Noisy, Unstable?
Hectagon (100-gon)	100	D100, Convex	(No specific associations)	Complexity, Diffusion, Overload	Very Low	Diffuse, Noisy, Unstable?

Note: This lexicon provides examples and is not exhaustive. Semantic load and potential generative effects are theoretical estimates.

II. Encoding Geometry in Generative Prompts: Tokenization and Latent Structures

Translating the desired geometric form and its associated symbolic intent into a format that generative models can interpret requires careful consideration of prompting strategies, AI tokenization processes, latent space representations, and potential points of failure.

2.1. Prompting Strategies for Geometric Control

Several approaches can be employed to specify polygonal forms in prompts:

- **Explicit Geometric Terms:** The most direct method involves naming the polygon, often with qualifiers for regularity or type (e.g., "a regular convex hexagon," "a blue pentagram," "a 24-sided polygonal sigil").³⁰ This relies heavily on the model's vocabulary and its learned association between the term and the corresponding visual concept.
- **Implicit Geometric Cues:** Geometry can be suggested indirectly through associated concepts or contexts. Prompts like "a pattern resembling a honeycomb" imply hexagons, "a stop sign shape" implies an octagon, or "a structure with perfect stability" might suggest squares.³⁰ Abstract concepts like "labyrinthine recursive harmony" could imply complex, high-n structures. This method leverages the model's associative knowledge but offers less precise control.
- **Structural/Relational Descriptions:** Instead of naming, one might describe the construction or properties (e.g., "a closed shape made of 7 equal straight lines meeting at equal angles," "a tiling composed of equilateral triangles"). This approach tests the model's ability to interpret geometric rules, which is often limited in standard T2I models.
- **Numerical Parameters:** Some interfaces or model architectures might allow direct numerical input for parameters like side count (e.g., using flags like --sides 12 or parameters within the prompt like n=20 polygon). The effectiveness depends entirely on whether the model is designed to parse and utilize such parameters.

2.2. AI Tokenization of Geometric Concepts

Generative models, particularly those using transformer architectures (common in LLMs and diffusion model encoders like CLIP), process input text by first breaking it into tokens.¹ This tokenization step is critical for geometric prompting:

- **Vocabulary Limitations:** Standard tokenizers are trained on vast corpora of natural language and image captions.³¹ Common geometric terms like "triangle," "square," and "circle" are likely well-represented and map to distinct tokens. However, less frequent terms, especially for higher-n polygons ("hendecagon," "triacontagon," "enneacontagon"), may not exist as single tokens in the model's vocabulary.²⁴
- **Subword Tokenization:** Many tokenizers use subword units (like Byte Pair Encoding or WordPiece). This means uncommon words are broken into smaller, more frequent pieces. "Enneacontagon" might be tokenized as "ennea", "##conta", "##gon". This fragmentation can dilute or obscure the specific meaning, particularly the crucial numerical information (90 sides).
- **Lack of Geometric Granularity:** The fundamental issue is that standard tokenization schemes are optimized for semantic meaning in natural language, not for the precise numerical and structural information inherent in geometry. A model might recognize "hexagon" (6 sides) and "octagon" (8 sides) distinctly, but treat "heptacontagon" (70 sides) and "octacontagon" (80 sides) as functionally equivalent tokens simply meaning "many-sided polygon" or even "circle." This loss of numerical specificity during the initial encoding stage severely hinders precise control over high-n shapes. This contrasts sharply with specialized approaches like Octree-based Adaptive Tokenization (OAT) for 3D shapes, which explicitly adapts token allocation based on geometric complexity³², or proposed methods like Geo-MAGVIT aimed at better diagram tokenization.³³ These specialized methods highlight the deficiencies of general-purpose tokenizers for geometric tasks.

2.3. Latent Space Representation of Geometry

The tokenized prompt embedding guides the generative process, typically by conditioning the latent space in

diffusion models.¹ The way geometry is represented in this high-dimensional space is crucial:

- **Feature Association:** Does the token for "hexagon" activate latent features corresponding specifically to six-sidedness, angularity, and D6 symmetry? Or does it activate broader features like 'angular', 'symmetrical', 'closed shape'? The specificity of this mapping influences rendering accuracy.
- **Structural vs. Semantic Encoding:** Models might prioritize semantic associations over structural accuracy. A prompt for an "octagonal stop sign" might yield a recognizable stop sign that is only vaguely octagonal, because the 'stop sign' concept is more strongly represented in the latent space than the precise 'octagon' geometry.
- **Implicit Structural Awareness:** Some models might implicitly learn geometric features through mechanisms like edge detection during training or employ separate conditioning channels for structural information, akin to transmitting edge maps alongside semantic descriptions in advanced semantic communication systems.³ This could potentially improve geometric fidelity independently of the main text prompt's tokenization.

2.4. Latent Instability and Semantic Collapse

Failures in geometric prompting often manifest when the model struggles to interpret or render the specified shape, particularly for high-n polygons:

- **Semantic Ignorance:** The model generates an image consistent with other parts of the prompt but completely ignores the geometric term. This suggests the token's influence on the latent space was too weak or overridden by other prompt elements.
- **Semantic Collapse:** The model interprets specific high-n terms (e.g., "triacontagon," "enneacontagon") not according to their precise definition but as a more general or visually similar concept. Common collapses include rendering a circle, a generic "complex blob," or simply noise. This indicates a failure to differentiate between high-n values in the latent representation, often stemming from the tokenization issues described earlier. Models may struggle even with basic polygon properties, like calculating diagonals for a decagon.³⁵
- **Rendering Instability/Incoherence:** Even if the model attempts to render a high-n shape, the process can become unstable, leading to visual artifacts, jagged or uneven sides, incorrect vertex connections (poor topology³⁶), excessive or nonsensical detail, or a breakdown into chaotic patterns. This reflects the difficulty of maintaining geometric consistency and precision through the iterative denoising process of diffusion models.¹

These failure modes are likely interconnected. An initial failure in tokenization (Section 2.2) can lead to a weak or inaccurate representation in the latent space (Section 2.3). If this latent signal is too weak or conflicts strongly with other prompt elements, semantic ignorance may result. If the model attempts to interpret a poorly defined or poorly differentiated high-n concept, semantic collapse or rendering instability are probable outcomes. The difficulty in generating topologically correct geometry is a known challenge³⁶, suggesting these failures are rooted in fundamental limitations of current generative architectures when dealing with precise geometric constraints.

(See Guide 1: Prompt Grammar Syntax for Polygonal Encoding at the end of this section)

Guide 1: Prompt Grammar Syntax for Polygonal Encoding

This guide provides recommendations for structuring prompts to enhance the likelihood of achieving desired polygonal forms in generative outputs, particularly visual ones. Effectiveness varies significantly between models and versions.

1. Explicit Naming:

- **Best Practice:** Be specific. Use "regular convex [polygon name]" (e.g., "regular convex octagon").
 - **Star Polygons:** Use Schläfli symbols if the model might recognize them (unlikely for most) or describe verbally: "a star polygon connecting every second vertex of a heptagon" for {7/2}.
 - **Qualifiers:** Add adjectives describing appearance: "a wireframe dodecahedron," "a solid gold hexagon."
2. **Numerical Specification (Model Dependent):**
- **Direct Parameters:** If supported, use flags or parameters: --sides 12, polygon(n=20). Check model documentation.
 - **In-Prompt Phrasing:** Try natural language specification: "a polygon with 15 equal sides," "an n-gon where n=30." Clarity can be low.
3. **Implicit Cues:**
- **Associated Objects:** "shape of a stop sign," "beehive cell pattern," "cross-section of a pencil."
 - **Associated Concepts:** "a structure embodying perfect stability" (square), "a symbol of natural efficiency" (hexagon). Less reliable for specific geometry.
4. **Compositional Role:**
- **Subject:** "A large, central enneactagon."
 - **Frame/Boundary:** "An image framed by an intricate octagonal border."
 - **Element:** "A shield bearing a pentagram symbol."
 - **Pattern/Tiling:** "A background tiled with alternating squares and triangles."
5. **Style/Attribute Modifiers:**
- Combine geometry with artistic style: "Art Deco octagonal mirror," "glowing neon triangle sign," "watercolor painting of a heptagon."³⁰
 - Specify material/texture: "polished chrome dodecahedron," "rough stone square block," "translucent glass pentagon."³⁰
6. **Handling High-n Polygons (Strategies to Mitigate Collapse):**
- **Emphasize Distinctness:** "a regular 50-gon with clearly defined, sharp corners and distinct flat sides, not a circle."
 - **Visual Analogies:** "a shape like a multi-faceted coin with 40 sides."
 - **Negative Prompting:** Use negative prompts to exclude undesired outcomes: --no circle, round, smooth.
 - **Iterative Refinement:** Generate an approximation, then use image-to-image techniques with prompts emphasizing correction: "refine the shape to have exactly 60 sides."
 - **Structural Priming:** Include related structural elements: "a regular icosagon inscribed within a visible circle."
 - **Accept Approximation:** For very high n, aiming for "a highly complex, many-sided polygon approximating a circle" might be more realistic than demanding perfect geometric accuracy.

Note: These are heuristics. Model interpretation of geometric terms is often inconsistent and imperfect, especially for complex or uncommon shapes. Experimentation is key.

III. Perceptual Thresholds: Cognitive Load and Aesthetic Harmony

The effectiveness of using polygons in generative systems is constrained not only by AI limitations but also by the perceptual and cognitive capacities of the human observer. How humans perceive complexity, symmetry, and harmony in shapes influences both the interpretation of generated outputs and the potential symbolic resonance.

3.1. Human Visual Perception of Polygonal Complexity

Research in visual perception and cognitive psychology has long investigated how humans process visual complexity, including geometric forms.¹² Key findings relevant to polygons include:

- **Metrics of Complexity:** Objective measures correlating with perceived complexity include the number of sides/angles, perimeter²/area ratio, deviations from regularity (variance in side lengths or angles), symmetry, and information content (redundancy).¹² Automated measures like edge detection ("Perimeter" metric) and image compression size (e.g., GIF compression) also correlate well with subjective complexity ratings, potentially offering more objective assessments less biased by familiarity.³⁹
- **Complexity, Redundancy, and Processing:** Studies using random vs. redundant (rule-generated) polygons reveal a trade-off.¹² More complex, less redundant (random) polygons contain more information and are easier to discriminate from one another in recognition tasks. However, simpler, more redundant polygons are easier to reproduce accurately after brief exposures, though harder to tell apart.¹² This suggests complexity aids distinctiveness, while simplicity/redundancy aids memorability or reconstruction.
- **Perceptual Flattening:** While not explicitly quantified for polygons in the provided material, human visual acuity and cognitive processing have limits. As the number of sides n of a regular polygon increases, it becomes progressively harder to distinguish individual sides and vertices. Beyond a certain threshold, the polygon becomes perceptually indistinguishable from a circle or is simply categorized as a "complex rounded shape." This "perceptual flattening" limits the effective range of n for conveying distinct polygonal identity to a human viewer.

3.2. AI "Perception" and Representation of Complexity

AI models, particularly diffusion models, do not "perceive" in the human sense, but they do process and represent complexity based on their architecture and training data.

- **Learning Low-Dimensional Distributions:** Diffusion models are theorized to generalize well by effectively learning low-dimensional distributions underlying high-dimensional data like images.³⁴ Real-world images, despite having many pixels, often have lower intrinsic dimensionality. This suggests models might be biased towards representing data on simpler manifolds. Such an inductive bias could make it inherently difficult for them to faithfully represent the fine-grained, high-frequency detail needed for geometrically accurate high- n polygons, potentially leading to a form of "representational flattening" where complex polygons are simplified or collapse towards smoother, lower-dimensional forms like circles.
- **Statistical vs. Structural Representation:** AI models primarily learn statistical correlations and patterns from data.⁴⁰ Their representation of a polygon might be based on learned visual features (e.g., typical angles, textures associated with polygonal objects in the training set) rather than an explicit, rule-based geometric construction. While models can be designed or trained for geometric accuracy³¹, standard large-scale models may lack this specific capability.

The limitations faced by humans and AI regarding complex polygons thus appear to stem from different mechanisms. Human limitations are rooted in biological sensory and cognitive processing constraints, leading to perceptual flattening. AI limitations may arise from architectural biases favoring lower-dimensional representations, insufficient or biased training data lacking diverse geometric examples, or fundamental failures in tokenizing and encoding precise geometric information (as discussed in Section II). Consequently, an AI might fail to render a high- n polygon accurately not because it cannot "see" the detail, but because its internal representation or generative process simplifies, distorts, or collapses the structure.

3.3. Cognitive Resonance and Aesthetic Preference

The way shapes are perceived influences aesthetic judgments and cognitive processing efficiency.

- **Gestalt Principles:** Principles like symmetry, closure, proximity, and common region describe how humans naturally organize visual elements into unified wholes.²³ Designs adhering to these principles are often processed more easily and perceived as more harmonious. Symmetry and order, in particular, are linked to perceptions of balance and completeness.²³
- **Complexity and Aesthetics:** The relationship is debated. Birkoff proposed a negative correlation: aesthetic value increases with order and decreases with complexity.¹³ Berlyne, however, proposed an inverted U-shape: stimuli with moderate complexity are preferred over very simple or very complex ones.¹³ This suggests that while simple polygons might be readily processed, increasing n could initially enhance visual interest, but beyond a certain point, excessive complexity might reduce aesthetic appeal due to increased cognitive load.
- **Processing Efficiency:** Eye-tracking studies indicate that simpler visual compositions, particularly those using the Gestalt principle of proximity, demand less viewing time, elicit fewer fixations, and receive higher aesthetic ratings.¹³ Conversely, more complex stimuli require more cognitive effort (more fixations, longer viewing times) and tend to be rated lower aesthetically.¹³ This implies that as polygonal complexity (n) increases, the cognitive load rises, potentially diminishing aesthetic resonance beyond a moderate threshold.

3.4. Symmetry and Perceived Balance

Regular polygons are inherently highly symmetric, possessing D_n symmetry.⁶ This property strongly connects to human perception of balance and order, key components of Gestalt psychology and aesthetic preference.¹³

The high degree of rotational and reflectional symmetry in regular polygons likely contributes positively to their perceived stability and harmony in generated images. This inherent geometric structure might also serve as a powerful constraint or prior during the AI generation process. In diffusion models, which iteratively refine noise, a strong symmetrical constraint from the prompt could guide the denoising steps towards more coherent and stable outputs, resisting the tendency towards noise or structural collapse often seen in complex generations.³⁷ This stabilizing effect might be most pronounced for low-to-mid-range polygons where the symmetry is visually distinct and easily recognizable. For very high- n polygons, although mathematically highly symmetric, the perceptual flattening towards a circle might diminish the effectiveness of symmetry as a distinct visual stabilizer, potentially allowing instability to creep back in as the model struggles with the sheer number of sides and vertices.

IV. Generative Fidelity and Failure Modes

Assessing the ability of current generative models to accurately render prompted polygons requires understanding evaluation methods, cataloging common failure types, identifying thresholds where fidelity breaks down, and comparing performance across different models.

4.1. Benchmarking Geometric Accuracy in Text-to-Image Models

Evaluating T2I models typically focuses on semantic alignment (does the image match the text?) and overall image quality (realism, aesthetics). However, assessing geometric fidelity requires more specialized approaches.⁴³ Challenges include:

- **Lack of Standard Metrics:** Metrics like FID or CLIP score do not specifically measure geometric accuracy. Evaluating shape fidelity might require metrics focused on structural similarity, edge map comparison, or

recognition of specific geometric primitives within the output.⁴¹

- **Subjectivity vs. Objectivity:** While human evaluation can judge visual correctness, objective, automated methods are needed for large-scale benchmarking. Techniques from computer vision for shape analysis or object detection might be adapted.⁴¹
- **Complexity of Geometry:** Models struggle not just with individual shapes but also with complex spatial relationships, topological correctness (e.g., ensuring a solid is watertight or a simple polygon doesn't self-intersect), and precise parameterization.³⁶ Research is ongoing to develop models with better geometric understanding, sometimes involving specialized training or architectures.³¹

4.2. A Typology of Polygonal Rendering Failures

Based on user experiences and analysis of model capabilities, common failures when prompting for polygons include:

- **Side Count Inaccuracy:** The most basic failure: generating a polygon with $n \pm k$ sides instead of the requested n . This is particularly common for less frequent polygons (e.g., requesting a heptagon but getting a hexagon or octagon).
- **Irregularity:** Failing to maintain the defining properties of regular polygons – equal side lengths and equal interior angles. Generated shapes often have uneven sides or distorted angles.
- **Corner Malformation:** Vertices may be rendered as rounded, blurred, or improperly joined, losing the sharpness expected of polygonal corners.
- **Topological Errors:** Simple polygons might be rendered with self-intersecting edges. Star polygons might have incorrect vertex connections or winding numbers. Maintaining correct topology is a known challenge.³⁶
- **Incoherence/Noise:** Especially with high- n prompts, the generation process can break down entirely, resulting in random noise, chaotic patterns, or overly detailed textures that obscure any underlying polygonal structure.³⁷
- **Semantic Drift/Substitution:** The model replaces the requested geometric primitive with a semantically related but geometrically distinct object. For example, prompting "octagon" might yield a stop sign, or "triangle" might yield a pyramid. This indicates the semantic concept overshadowed the geometric instruction in the latent space.

4.3. Thresholds for Visual Drift and Symbolic Breakdown

The prevalence of these failures appears correlated with the complexity (n) of the requested polygon:

- **Low- n (approx. 3-6):** Models generally achieve higher fidelity for triangles, squares, pentagons, and hexagons. Failures are more often related to achieving perfect regularity (slight distortions in angles or sides) or integrating the shape naturally into a complex scene. Side count accuracy is usually good.
- **Mid- n (approx. 7-20):** Fidelity decreases. Side count errors become more frequent. Maintaining regularity is challenging. Models may struggle significantly with less common shapes in this range (e.g., hendecagon (11-gon), which is also non-constructible with compass and straightedge²⁵). The specificity of the shape starts to blur.
- **High- n (approx. >20):** This range sees a high probability of significant failure. Semantic collapse is common, with models defaulting to rendering circles, near-circles, or generic "complex shapes" regardless of the specific n requested (e.g., for a triacontagon (30-gon) or enneactontagon (90-gon)). Rendering instability, incoherence, noise, and gross geometric inaccuracies (like uneven sides, wildly incorrect angles) are prevalent. Attempts to render distinct sides often fail, resulting in lumpy or wavy approximations of a circle. The model's internal representation seems unable to differentiate or maintain the structure for such high side counts, likely

due to the tokenization and representational limits discussed previously. Even basic properties of simpler polygons like decagons can be misrepresented.³⁵

These thresholds suggest a practical limit to the effective use of explicit high-n polygon prompts in current general-purpose T2I models.

4.4. Model-Specific Performance (Midjourney vs. SDXL vs. Others)

Direct, systematic benchmark comparisons of polygon rendering fidelity across major T2I models are scarce in publicly available research. However, inferences can be drawn from model capabilities and user experiences:

- **Midjourney:** Known for high aesthetic quality and coherence. Its interpretation of prompts, including geometric elements, is often effective for common shapes integrated into artistic scenes. However, precise geometric control, especially for high-n or non-standard polygons, can be challenging, as the model prioritizes overall image plausibility and style.³⁰
- **Stable Diffusion (SD) / SDXL:** Offers more open architectures and fine-tuning possibilities (e.g., via ComfyUI).⁴⁴ SDXL incorporates architectural improvements like a larger UNet backbone and a second text encoder, designed to capture more nuanced detail.³¹ This might translate to potentially better handling of complex prompts, including geometric specifications, compared to earlier SD versions. However, base models still struggle with precise geometry without specialized fine-tuning or control mechanisms (like ControlNet). User reports suggest T2I alone often fails to achieve precise image descriptions.⁴⁴
- **DALL-E 3 (via ChatGPT/Bing):** Integrates with large language models for potentially better prompt understanding and decomposition. This might aid in interpreting more complex geometric descriptions, although the underlying image generation model still faces the same core challenges in rendering precision.
- **Specialized Models:** Research is exploring models or techniques specifically aimed at improving geometric fidelity, such as incorporating normal map estimation or using geometry-aware training paradigms.³¹ These are not yet standard in widely accessible T2I platforms.

Differences in performance likely arise from a combination of factors. Architectural choices, such as the size and design of the UNet or the sophistication of the text encoder(s) (e.g., SDXL's enhancements³¹), play a role in the model's capacity to handle detail and complex instructions. Furthermore, the composition and quality of the training data are critical.⁴⁵ Models trained on datasets with more technical diagrams, architectural drawings, or explicitly labeled geometric shapes might exhibit better geometric fidelity than those trained primarily on natural images and artistic illustrations. Newer models may also incorporate specific techniques learned from research aimed at improving structural or geometric accuracy.⁴²

(See Table 2: Failure Heatmap and Table 3: Model Comparison for Polygonal Fidelity at the end of this section)

Table 2: Failure Heatmap for Regular Polygon Generation in T2I Models (Estimated Prevalence)

Polygon Complexity (n)	Side Count Error	Irregularity (Sides/Angles)	Corner Malformation	Topology Error (Simple/Star)	Incoherence / Noise	Semantic Collapse (to Circle/Blob)
3-6	Low	Medium	Low-Medium	Low	Low	Very Low
7-12	Medium	Medium-High	Medium	Low-Medium	Low-Medium	Low

13-20	Medium-High	High	Medium-High	Medium	Medium	Medium
21-50	High	High	High	Medium-High	Medium-High	High
>50	Very High	Very High	High	High	High	Very High

Note: Prevalence (Low/Medium/High/Very High) is an estimated average across common T2I models like Midjourney, SDXL. Actual results vary significantly.

Table 3: Model Comparison for Polygonal Fidelity (Qualitative Assessment)

Model (Representative Version)	Strengths in Geometric Rendering	Weaknesses / Common Failures	Specific Control Mechanisms (Potential)	Overall Fidelity Rating (Low / Mid / High-n)
Midjourney (e.g., v6)	Good integration of common shapes into artistic styles; high coherence.	Struggles with precise regularity, high-n accuracy, complex topology; prioritizes aesthetics.	Limited explicit geometric controls.	Good / Fair / Poor
SDXL 1.0 (Base)	Larger model capacity may handle detail better than older SD. ³¹	Base model still prone to irregularity, side count errors, high-n collapse; T2I often imprecise. ⁴⁴	Open architecture allows ControlNet, LoRAs, etc. for potential fine-tuning.	Fair-Good / Fair / Poor-Fair
DALL-E 3 (via ChatGPT)	Potentially better prompt understanding via LLM integration.	Underlying generation faces similar fidelity issues; high-n accuracy remains challenging.	Relies on LLM interpretation of text.	Fair-Good / Fair / Poor-Fair
Specialized/Research Models	May incorporate geometry-specific training or architectures. ⁴²	Often not publicly available or easy to use; may be domain-specific.	Varies (e.g., normal map conditioning).	Potentially Excellent / Good / Fair (Targeted)

Note: Ratings are qualitative estimates based on available information and general model characteristics. Performance is highly prompt-dependent and evolves rapidly.

V. Cross-Modality Transfer: Geometry Beyond the Visual

While the visual rendering of polygons presents significant challenges, their structural properties—particularly side count (n) and symmetry—might be transferable to other modalities like music and motion, potentially offering more robust control, especially for high-n values.

5.1. Theoretical Basis for Cross-Modal Mapping

Modern generative AI increasingly operates across multiple modalities, with models capable of processing and

generating combinations of text, images, audio, and motion data.⁴ This relies on finding or learning shared representations or ways to translate between modality-specific encodings. For geometric shapes, the transfer likely focuses on abstract, quantifiable properties rather than direct visual appearance:

- **Number (n):** The side count is a discrete numerical value easily mapped to countable events or parameters in other domains (e.g., beats, notes, steps, poses).
- **Symmetry (Dn):** The mathematical structure of symmetry can be mapped to structural patterns in time (music) or space (motion).
- **Periodicity/Repetition:** The repetitive nature of sides and angles in a regular polygon can map to rhythmic or cyclical patterns.

5.2. Text-to-Music: Polygonal Rhythms and Structures

Mapping polygonal properties to music offers several possibilities, leveraging generative music frameworks that can process symbolic or textual input⁴:

- **Side Count (n) → Rhythm/Meter:** The number of sides could define the number of beats in a measure (e.g., $n=5$ for 5/4 time), the subdivision of beats (e.g., n -tuplets), or the length of rhythmic patterns.
- **Side Count (n) → Pitch/Harmony:** n could determine the number of unique pitches in a scale or mode, the number of notes in a recurring chord voicing, or the number of steps in a harmonic sequence or cycle.
- **Symmetry (Dn) → Formal Structure:** The reflectional and rotational symmetries could inspire musical forms. For example, reflectional symmetry could map to palindromic musical structures (playing forwards then backwards). Rotational symmetry could map to repeating sections or cyclical forms. Algorithmic art often involves mapping symbolic representations to outputs.¹⁰

5.3. Text-to-Motion: Polygonal Paths and Kinetics

Similarly, geometric properties can inform the generation of motion or animation, using text-to-motion models⁴⁹ or generative design principles¹¹:

- **Vertices → Key Poses/Waypoints:** The n vertices of a polygon could define n key poses in an animation sequence or n target locations in a spatial trajectory.
- **Sides → Movement Segments:** The n sides could represent n distinct segments of motion, perhaps with characteristics (speed, curvature) modulated by the prompt.
- **Overall Shape → Path Envelope:** The polygon itself could define a boundary or path that constrains the movement of an object or character.
- **Symmetry (Dn) → Choreographic Structure:** Symmetrical movements, formations involving multiple actors, or repetitive kinetic patterns could be derived from the polygon's symmetry group. Generative design sometimes uses natural forces or data (like sound) to create kinetic forms, suggesting abstract mappings are feasible.¹¹

5.4. Comparative Stability and Symbolic Clarity

A key question is whether non-visual modalities can handle high- n polygon prompts with greater stability and symbolic clarity than text-to-image models.

There is a basis to hypothesize that music and motion generation might be more robust in interpreting the *abstract numerical aspect* of high- n polygons. Mapping $n=90$ to "90 beats per minute" or "a sequence of 90 steps" is computationally far simpler and less ambiguous than accurately rendering a regular 90-sided figure with distinct

sides and correct angles. The visual modality struggles with the high spatial frequency and precise topology required for high- n rendering, leading to collapse or noise (Section IV). Temporal modalities (music, motion) primarily deal with sequencing and counting.

Therefore, while the visual representation might fail, the core symbolic intent related to *number* or *high complexity* could potentially be transferred more reliably to music or motion. A generated piece with a 90-beat rhythmic cycle might successfully convey the intended sense of "manyness" or "complexity" derived from an enneactagon prompt, even if a visual model fails to draw the shape. Symbolic clarity in these cases depends on the intuitiveness and consistency of the defined mapping (e.g., clearly stating "map sides to beats"). The challenge shifts from rendering fidelity to designing meaningful cross-modal correspondences.

VI. Synthesis and Reflexive Insights

The exploration of polygonal cognition in generative systems reveals a complex interplay between geometry, symbolism, computation, and perception. Synthesizing the findings allows for reflection on the core inquiries posed initially.

6.1. Complexity vs. Noise: The Polygonal Potency Paradox

Does increasing polygonal complexity (higher n) enhance symbolic potential or dilute it into noise? The analysis suggests a paradox. For low- n polygons (roughly $n \leq 12$), increasing complexity introduces richer, more specific geometric features and taps into established, albeit varied, symbolic associations.⁷ This offers significant symbolic potential for prompt control, provided the model can navigate the potential ambiguity (high semantic load).

However, beyond a certain threshold (around $n \approx 20-30$), the symbolic specificity rapidly diminishes.²⁴ High- n polygons primarily signify 'complexity' itself or 'approximation of a circle'.⁶ Simultaneously, generative fidelity plummets due to tokenization limits, representational flattening, and rendering instability (Section IV, Table 2). Thus, while mathematical complexity increases with n , the *effective symbolic potency* for precise communication peaks in the low-to-mid- n range and then declines. Pushing n too high risks not only diluting specific meaning but also introducing generative noise and failure, effectively reducing, rather than increasing, meaningful symbolic control. The inverted U-shape hypothesis for generative stability (Section 1.4), mirroring aesthetic preference curves¹³, appears plausible.

6.2. The Limits of Precision in Geometric Prompting

At what point does precise geometric prompting become futile? The evidence indicates that for current general-purpose T2I models, specifying high- n regular polygons (e.g., "regular enneactagon") with the expectation of accurate rendering is largely ineffective. The failure cascade—poor tokenization of rare terms, leading to weak or collapsed latent representations, resulting in inaccurate or noisy output (Section 2.4)—means the model likely cannot interpret or execute such precise instructions.

This highlights a crucial trade-off between prompt precision and generative robustness. While explicit naming works reasonably well for common low- n shapes, implicit prompting (using associated concepts) or aiming for approximation might be more pragmatic strategies for higher complexity. The pursuit of absolute geometric precision may require specialized models or control mechanisms (e.g., ControlNet, geometry-focused architectures⁴²) rather than relying solely on natural language prompts interpreted by standard T2I systems. The inherent limitations in representing complex topology and parameters remain a barrier.³⁶

6.3. Emergent vs. Projected Symbolism

Are polygonal symbolic associations merely human projections encoded in training data, or do geometric properties emergently afford certain meanings? The analysis suggests both factors are at play. Many specific associations (e.g., triangle/trinity, pentagram/occultism) are clearly culturally constructed and projected onto the shapes; AI models learn these correlations from text and image data.⁸

However, some fundamental associations seem grounded in perceptual and cognitive principles (Gestalt psychology¹³). The symmetry (D_n) of regular polygons inherently affords perceptions of balance, order, and stability.⁶ The sharp points of a triangle afford dynamism or threat. The enclosure of any polygon affords concepts of containment or definition. These affordances might represent a more universal, perhaps emergent, layer of meaning derived from the interaction of geometric structure with cognitive processing mechanisms.

Generative AI complicates this picture. Models operate as 'black boxes', learning complex statistical relationships.⁴⁰ It is difficult to ascertain whether an AI associating squares with 'stability' is merely replicating training data correlations or if its internal representations have captured some deeper structural property related to human perception of stability. The meaning generated by the AI is ultimately a product of its training data and architecture, potentially blending projected cultural meanings with patterns reflecting inherent structural affordances.

6.4. Implications and Future Directions

This investigation carries implications for several fields:

- **Prompt Engineering:** Requires awareness of model limitations regarding geometric terms, especially high- n polygons. Strategies should adapt, using explicit terms for low- n , implicit cues, or approximation goals for high- n , and leveraging negative prompting or iterative refinement (Guide 1). Understanding semantic load and potential failure modes (Table 2) is crucial.
- **UX-AI Design:** Interfaces for creative AI tools could benefit from incorporating explicit geometric controls beyond text prompts, perhaps visual manipulators or parameter sliders, acknowledging the limitations of purely linguistic specification.
- **Symbolic Systems & Cognitive Science:** Generative AI provides a new testbed for exploring how symbolic meaning is represented, processed, and potentially transformed within non-human cognitive architectures. The failures and successes in rendering symbolic geometry offer insights into the models' internal representations and limitations.

Future research should prioritize several areas:

- **Geometry-Aware Models:** Developing and evaluating tokenizers and model architectures specifically designed for geometric understanding and precision, building on ideas like OAT³² or aiming for better diagrammatic reasoning.³³
- **Systematic Benchmarking:** Establishing rigorous benchmarks and metrics specifically for evaluating geometric fidelity (polygon accuracy, topology, parameterization) across different T2I models.⁴³
- **Exploring Variations:** Extending the analysis beyond regular convex polygons to include irregular polygons, star polygons⁶, and potentially 3D polyhedra.
- **Cross-Modal Mapping:** Conducting deeper empirical studies on the stability and semantic clarity of transferring polygonal properties (n , symmetry) to music, motion, and other modalities, testing the hypothesis that abstract properties transfer more robustly than visual form.
- **Investigating Prompt Strategies:** Systematically testing the effectiveness of the proposed "Prompt Upgrades":

- *Symbol-Shape Inversion*: Prompting for high- n polygons intended to evoke minimalism could reveal how models handle contradictory semantic and structural cues.
- *Polygonal Grammar Composer*: Developing a modular prompt language where n explicitly controls parameters like tone, rhythm, or spatial hierarchy would test the feasibility of direct numerical control.
- *Prompt Drift Analyzer*: Tracing how the output for a specific polygon prompt (e.g., "heptagon") varies across different models (Midjourney, SDXL, DALL-E 3) could quantify model-specific biases and interpretations.

Conclusion: Recapitulating the Role of Semantic Geometry in AI Composition

Regular polygons, as fundamental geometric primitives, offer a potentially rich vocabulary for influencing generative AI outputs. Their inherent structural properties, particularly side count (n) and symmetry (D_n), combined with accrued cultural and psychological symbolism, provide affordances for controlling composition, aesthetics, and meaning. Low- n polygons carry significant, albeit complex and culturally variable, semantic load, enabling symbolic prompting for concepts like stability (square), dynamism (triangle), efficiency (hexagon), or completion (decagon).

However, the practical application of this "semantic geometry" is currently constrained by significant limitations in both AI models and human perception. As polygonal complexity (n) increases, specific symbolic associations fade, generative fidelity decreases sharply due to issues in tokenization, latent representation, and rendering stability, and human perception eventually flattens the distinct shape towards a circle. Consequently, prompting for high- n polygons with expectations of precise geometric rendering or specific symbolic meaning beyond "complexity" is often futile in current general-purpose models.

The potential for cross-modal transfer, mapping abstract properties like side count or symmetry to music or motion, offers a promising avenue where the numerical essence of polygons might be conveyed more robustly than their visual form. Ultimately, harnessing the full potential of polygonal cognition in generative systems requires advancements in AI architectures towards better geometric understanding, more sophisticated prompt engineering strategies aware of current limitations, and potentially hybrid interfaces combining linguistic, numerical, and visual controls. Understanding the interplay between the formal language of geometry, the fluid language of symbols, the statistical language of AI, and the perceptual language of humans remains key to shaping a future where generative systems can be guided with greater precision, nuance, and semantic depth.

Works cited

1. On the Challenges and Opportunities in Generative AI - arXiv, accessed May 12, 2025, <https://arxiv.org/html/2403.00025v3/>
2. Generative artificial intelligence: a historical perspective - PMC, accessed May 12, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11970245/>
3. Generative AI for Semantic Communication: Architecture ..., accessed May 12, 2025, https://www.researchgate.net/publication/388738636_Generative_AI_for_Semantic_Communication_Architecture_Challenges_and_Outlook
4. UniMuMo: Unified Text, Music and Motion Generation - arXiv, accessed May 12, 2025, <https://arxiv.org/html/2410.04534v1>
5. Geometric primitive - Wikipedia, accessed May 12, 2025, https://en.wikipedia.org/wiki/Geometric_primitive

6. Regular polygon - Wikipedia, accessed May 12, 2025, https://en.wikipedia.org/wiki/Regular_polygon
7. A Guide to Psychology of Shapes in Design - Pixcap, accessed May 12, 2025, <https://pixcap.com/blog/psychology-of-shapes>
8. Semiotics - Wikipedia, accessed May 12, 2025, <https://en.wikipedia.org/wiki/Semiotics>
9. Semiotic schemas: A framework for grounding language in action and perception - Social Machines, accessed May 12, 2025, https://ism.media.mit.edu/papers/semiotic_schemas_2005.pdf
10. About Algorithmic Art - Radical Art, accessed May 12, 2025, <http://radicalart.info/AlgorithmicArt/intro.html>
11. page 373 GA2015 - XVIII Generative Art Conference Pinar Calisir NATURAL PERIODIC FOR, accessed May 12, 2025, http://generativeart.com/ga2015_WEB/periodic-forces_Calisir.pdf
12. (PDF) Visual Complexity: A Review - ResearchGate, accessed May 12, 2025, https://www.researchgate.net/publication/7337589_Visual_Complexity_A_Review
13. Influence of Complexity and Gestalt Principles on Aesthetic ..., accessed May 12, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11379339/>
14. Star polygon - Wikipedia, accessed May 12, 2025, https://en.wikipedia.org/wiki/Star_polygon
15. Shape, Definition, Properties, Examples | Regular Polygons - Meaning - Cuemath, accessed May 12, 2025, <https://www.cuemath.com/geometry/polygons/>
16. The Psychological Meanings Behind Familiar Shapes (and How to Use Them) - Shutterstock, accessed May 12, 2025, <https://www.shutterstock.com/blog/psychological-meaning-shapes-use>
17. The History and Symbolism of the Pentagon - Dual Crossroads, accessed May 12, 2025, <https://www.dualcrossroads.com/post/the-history-and-symbolism-of-the-pentagram>
18. Was the U.S. Pentagon inspired by ancient monumental architecture? - Ancient Origins, accessed May 12, 2025, <https://www.ancient-origins.net/ancient-places-americas-opinion-guest-authors/was-us-pentagon-ancient-monumental-102090>
19. Hexagon - Wikipedia, accessed May 12, 2025, <https://en.wikipedia.org/wiki/Hexagon>
20. Octagon - Wikipedia, accessed May 12, 2025, <https://en.wikipedia.org/wiki/Octagon>
21. Geometry - Decagon - Definition, Examples & Practice Problems - Bytelearn, accessed May 12, 2025, <https://www.bytelearn.com/math-topics/geometry-decagon>
22. Decagon: Properties, Construction, History, and Applications - Testbook, accessed May 12, 2025, <https://testbook.com/maths/decagon>
23. What are the Gestalt Principles? | IxDF - The Interaction Design Foundation, accessed May 12, 2025, <https://www.interaction-design.org/literature/topics/gestalt-principles>

24. enneacontagon - Wiktionary, the free dictionary, accessed May 12, 2025, <https://en.wiktionary.org/wiki/enneacontagon>
25. Definition of Polygons - Department of Mathematics at UTSA, accessed May 12, 2025, https://mathresearch.utsa.edu/wiki/index.php?title=Definition_of_Polygons
26. A Contemporary Survey on Semantic Communications: Theory of Mind, Generative AI, and Deep Joint Source-Channel Coding - arXiv, accessed May 12, 2025, <https://arxiv.org/html/2502.16468v1>
27. Examining cultural drifts in artworks through history and development: cultural comparisons between Japanese and western landscape paintings and drawings - PubMed Central, accessed May 12, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4168670/>
28. 2.3 Semiotic analysis of folklore and visual culture - Fiveable, accessed May 12, 2025, <https://library.fiveable.me/folklore-and-visual-culture/unit-2/semiotic-analysis-folklore-visual-culture/study-guide/kSWfdmAI1TLtCjLu>
29. What is Sacred Geometry? - Karien Bredenkamp, accessed May 12, 2025, <https://www.karienbredenkamp.com/blogs/learn/what-is-sacred-geometry>
30. Midjourney Prompts: A Guide for Beginners and Experts, accessed May 12, 2025, <http://manypixels.co/blog/graphic-design/midjourney-prompts>
31. A Generative Approach to High Fidelity 3D Reconstruction from Text Data - arXiv, accessed May 12, 2025, <https://arxiv.org/html/2503.03664>
32. (PDF) Efficient Autoregressive Shape Generation via Octree-Based ..., accessed May 12, 2025, https://www.researchgate.net/publication/390468151_Efficient_Autoregressive_Shape_Generation_via_Octree-Based_Adaptive_Tokenization
33. GeoUni: A Unified Model for Generating Geometry Diagrams, Problems and Problem Solutions - arXiv, accessed May 12, 2025, <https://arxiv.org/html/2504.10146v1>
34. Generalization of Diffusion Models: Principles, Theory, and ..., accessed May 12, 2025, <https://www.siam.org/publications/siam-news/articles/generalization-of-diffusion-models-principles-theory-and-implications/>
35. International Journal of Interactive Multimedia and Artificial Intelligence, accessed May 12, 2025, https://www.ijimai.org/journal/sites/default/files/journals/ijimai20248_5.pdf
36. BrepGen: A B-rep Generative Diffusion Model with Structured Latent Geometry, accessed May 12, 2025, <https://damassets.autodesk.net/content/dam/autodesk/www/pdfs/brepgen.pdf>
37. Generative Physical AI in Vision: A Survey - arXiv, accessed May 12, 2025, <https://arxiv.org/html/2501.10928v2>
38. Generative Adversarial Networks with Limited Data: A Survey and Benchmarking - arXiv, accessed May 12, 2025, <https://arxiv.org/html/2504.05456v1>
39. (PDF) Visual Complexity: Is That All There Is? - ResearchGate, accessed May 12, 2025, https://www.researchgate.net/publication/221097715_Visual_Complexity_Is_That

All_There_Is

40. What Is an AI Model? - IBM, accessed May 12, 2025,
<https://www.ibm.com/think/topics/ai-model>
41. A geometric neural solving method based on a diagram text information fusion analysis, accessed May 12, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11685888/>
42. Hi3DGen: High-fidelity 3D Geometry Generation from Images via Normal Bridging - GitHub Pages, accessed May 12, 2025,
https://stable-x.github.io/Hi3DGen/hi3dgen_paper.pdf
43. Challenges in Generating Accurate Text in Images: A Benchmark for Text-to-Image Models on Specialized Content - MDPI, accessed May 12, 2025,
<https://www.mdpi.com/2076-3417/15/5/2274>
44. As stable diffusion evolves are we in fact moving away from 'text to image' ? : r/StableDiffusion - Reddit, accessed May 12, 2025,
https://www.reddit.com/r/StableDiffusion/comments/1526u2m/as_stable_diffusion_evolves_are_we_in_fact_moving/
45. Why many generative AI projects might fail after the POC phase - DataGOL, accessed May 12, 2025,
<https://www.datagol.ai/blog/generative-ai-projects-might-fail>
46. Generative AI Projects Fail Amid High Costs and Risks - TechRepublic, accessed May 12, 2025,
<https://www.techrepublic.com/article/30-generative-ai-projects-fail-amid-high-costs-and-risks/>
47. Sounds Like Music: Toward a Multi-Modal Media Theory of Gaussian Pop, accessed May 12, 2025,
<https://www.cyberneticforests.com/news/toward-a-multi-modal-media-theory>
48. Artificial Intelligence in Music, Sound, Art and Design | springerprofessional.de, accessed May 12, 2025,
<https://www.springerprofessional.de/artificial-intelligence-in-music-sound-art-and-design/50899824>
49. Text to Blind Motion - PMC, accessed May 12, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12013345/>
50. (PDF) PersonaBooth: Personalized Text-to-Motion Generation - ResearchGate, accessed May 12, 2025,
https://www.researchgate.net/publication/389749433_PersonaBooth_Personalized_Text-to-Motion_Generation
51. Generate vector shape fills using text prompts in Illustrator - Adobe Help Center, accessed May 12, 2025,
<https://helpx.adobe.com/illustrator/using/generative-shape-fill.html>